Lecture 8 (10/2)

Correlation (2.4), Estimators, Bias, and Consistency

Standardized variables

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Consider the standardized variables

$$Z_X = (X - \mu_X)/\sigma_X, \quad Z_Y = (Y - \mu_Y)/\sigma_Y$$

What is the regression coefficient of Z_Y on Z_X?

Standardized variables

$$Z_Y = (Y - \mu_Y)/\sigma_Y$$

Standardized variables

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Z_Y &= (Y - \mu_Y)/\sigma_Y \\
&= \left[(\beta_0 + \beta_1 X + \epsilon) - (\beta_0 + \beta_1 \mu_X)\right]/\sigma_Y \quad \text{(substitute } Y, \mu_Y\text{)}
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$$

Therefore

$$\boxed{\beta_1 \sigma_X / \sigma_Y = \frac{\mathrm{cov}(X, Y)}{\sigma_X \sigma_Y}}$$

This is called the correlation coefficient and denoted ρ.

Coefficient of determination

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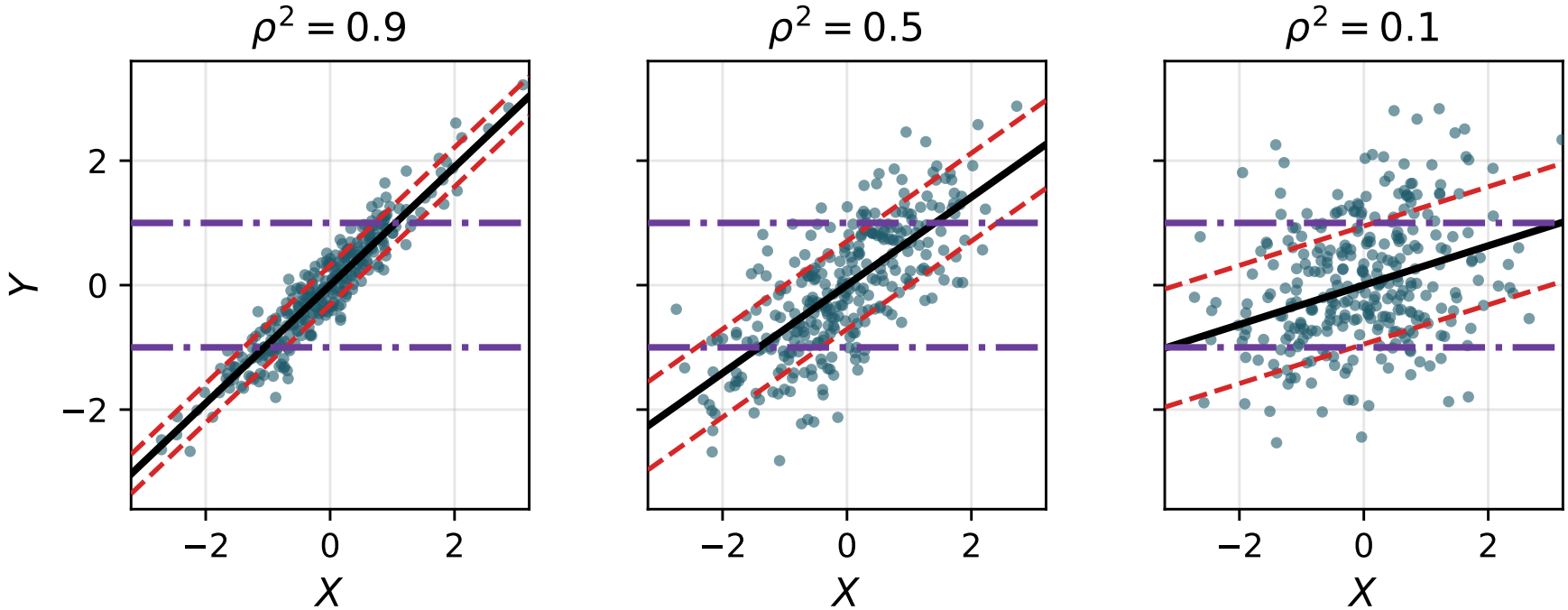
$$1 = \rho^2 + \frac{\sigma_\epsilon^2}{\sigma_Y^2} \qquad (\rho = \beta_1 \sigma_X / \sigma_Y)$$

Therefore

$$\boxed{1 - \rho^2 = \frac{\sigma_\epsilon^2}{\sigma_Y^2} = \frac{\text{var}(Y \mid X)}{\text{var}(Y)}}$$

ρ^2 is called the coefficient of determination

Visualizing ρ^2



Here $\mu_X = \mu_Y = 0$ and $\sigma_X = \sigma_Y = 1$ (hence $\mathrm{cov}(X, Y) = \beta_1 = \rho$). The black lines show $E[Y \mid X] = \rho X$ and red lines $\pm\sigma_\epsilon = \pm\sqrt{1 - \rho^2}$. Purple: $\pm\sigma_Y = \pm 1$.

Summary: measures of association

	Formula	Units	Range
$\text{cov}(X, Y)$		units Y $\times$ units X	$(-\infty, \infty)$
β_1	$\dfrac{\text{cov}(X, Y)}{\sigma_X^2}$	units Y/units X	$(-\infty, \infty)$
ρ	$\dfrac{\text{cov}(X, Y)}{\sigma_X \sigma_Y}$	none	$[-1, 1]$
ρ^2	$1 - \dfrac{\sigma_\epsilon^2}{\sigma_Y^2}$	none	$[0, 1]$

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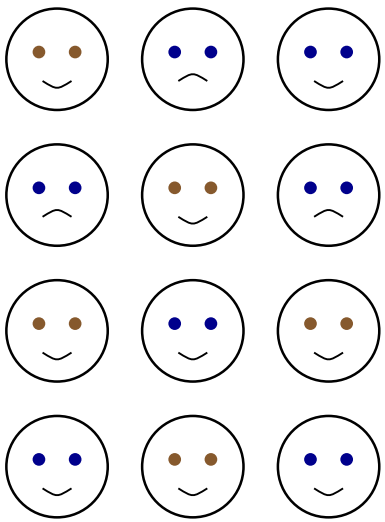
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- $\hat{\theta}$ is random – it will vary depending on which sample we collect.
- θ is a fixed, non-random number (in frequentist statistics; we will change this in Unit 6).

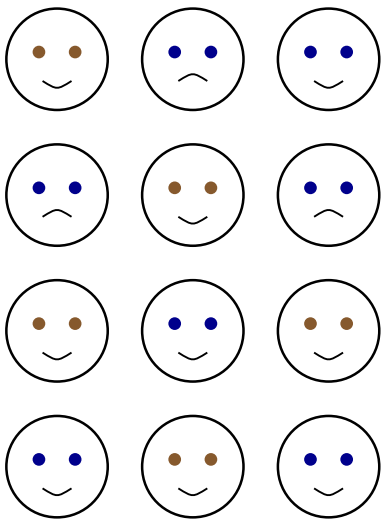
Replicates



Replicate 1

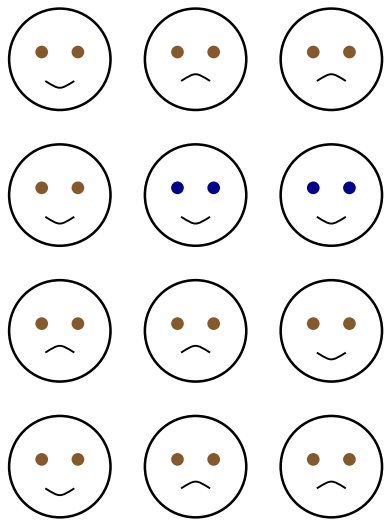
$\hat{q} = 3/12$

Replicates



Replicate 1

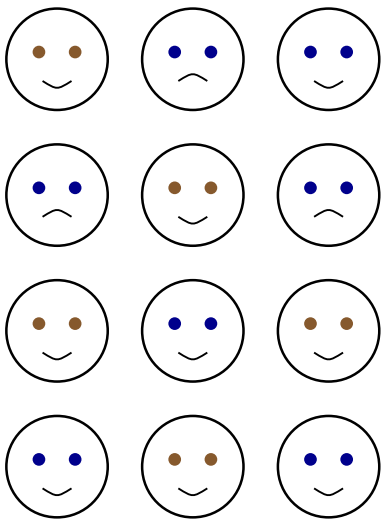
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Replicate 2

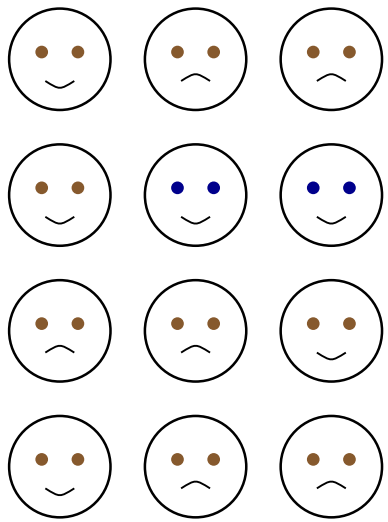
$\hat{q} = 6/12$

Replicates



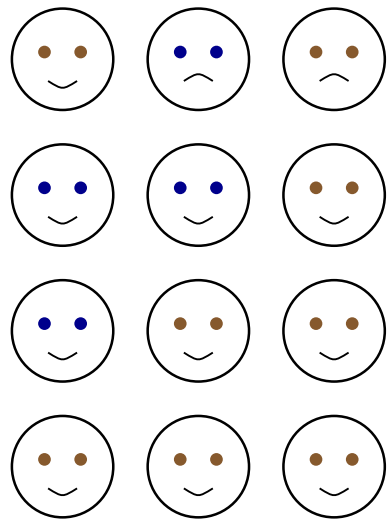
Replicate 1

$\hat{q} = 3/12$



Replicate 2

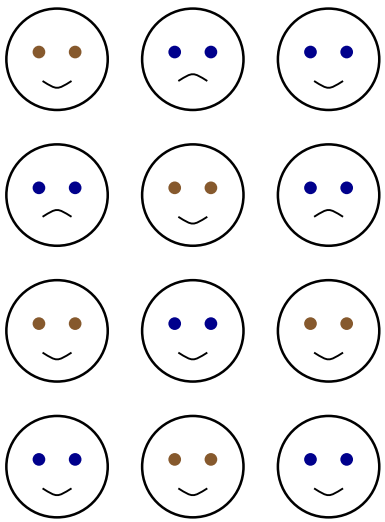
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Replicate 3

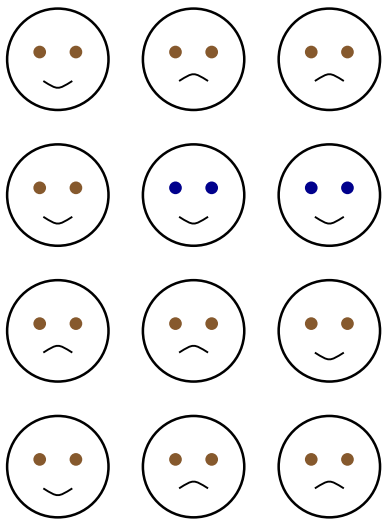
$\hat{q} = 2/12$

Replicates



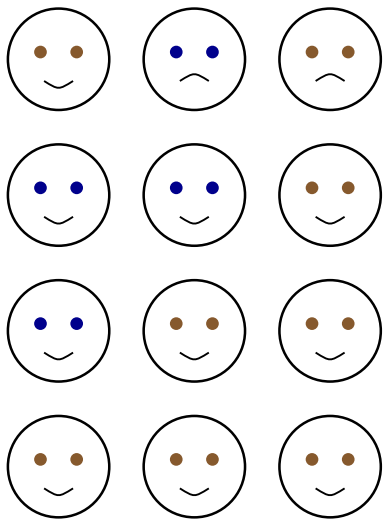
Replicate 1

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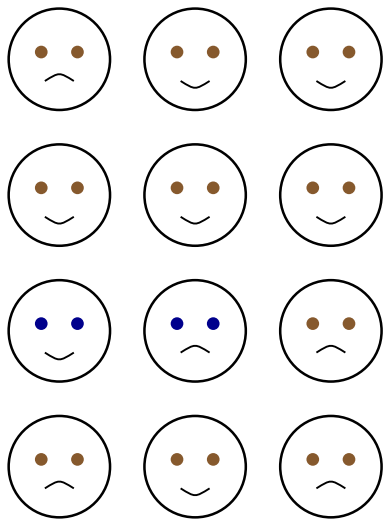
Replicate 2

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Replicate 3

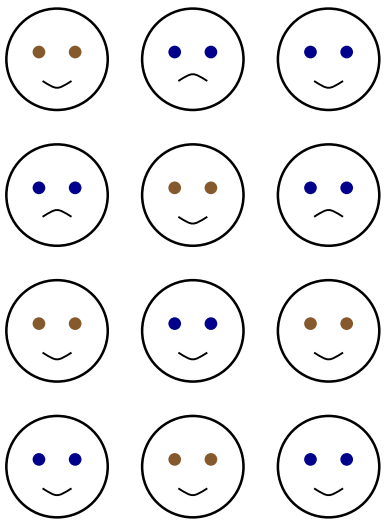
$\hat{q} = 2/12$



Replicate 4

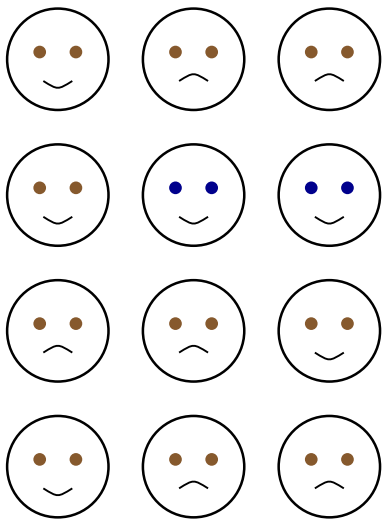
$\hat{q} = 5/12$

Replicates



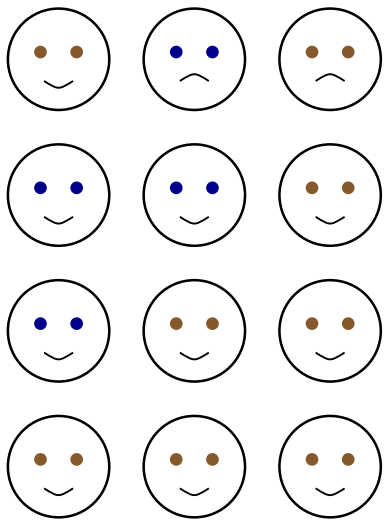
Replicate 1

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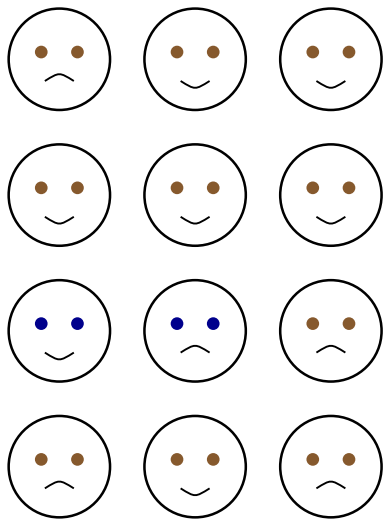
Replicate 2

$\hat{q} = 6/12$



Replicate 3

$\hat{q} = 2/12$



Replicate 4

$\hat{q} = 5/12$

$\hat{q}$ changes from replicate to replicate, while q stays fixed. The distribution of $\hat{q}$ over (infinitely many) replicates is its sample distribution; its spread is $\text{se}(\hat{q})$.

Sample distribution and standard error

Example

Let $Y_1, \ldots, Y_N \sim \text{Normal}(\mu, \sigma^2)$ and estimate μ with $\hat{\mu} = \overline{Y} = \frac{1}{N} \sum_i Y_i$.

Sample distribution and standard error

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A sum of Normals is Normal, so we know the distribution of $\hat{\mu}$ exactly:

$$\hat{\mu} \sim \text{Normal}\left(\mu, \ \frac{\sigma^2}{N}\right).$$

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The sample distribution is the distribution of $\hat{\mu}$ over replicates of the data. The standard deviation of the sample distribution is the standard error

$$\text{se}(\hat{\mu}) = \frac{\sigma}{\sqrt{N}}.$$

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Note: We need to remember that in reality we don't know σ, so this itself needs to be estimated. Some people call the distribution with estimated σ the sample distribution.

Estimating q

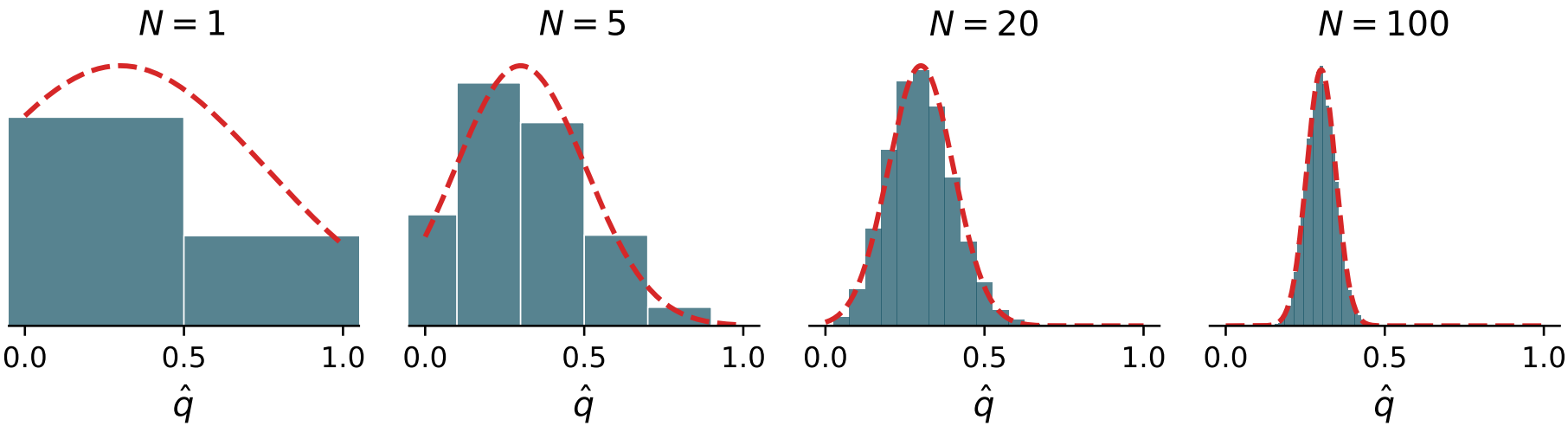
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Estimating q

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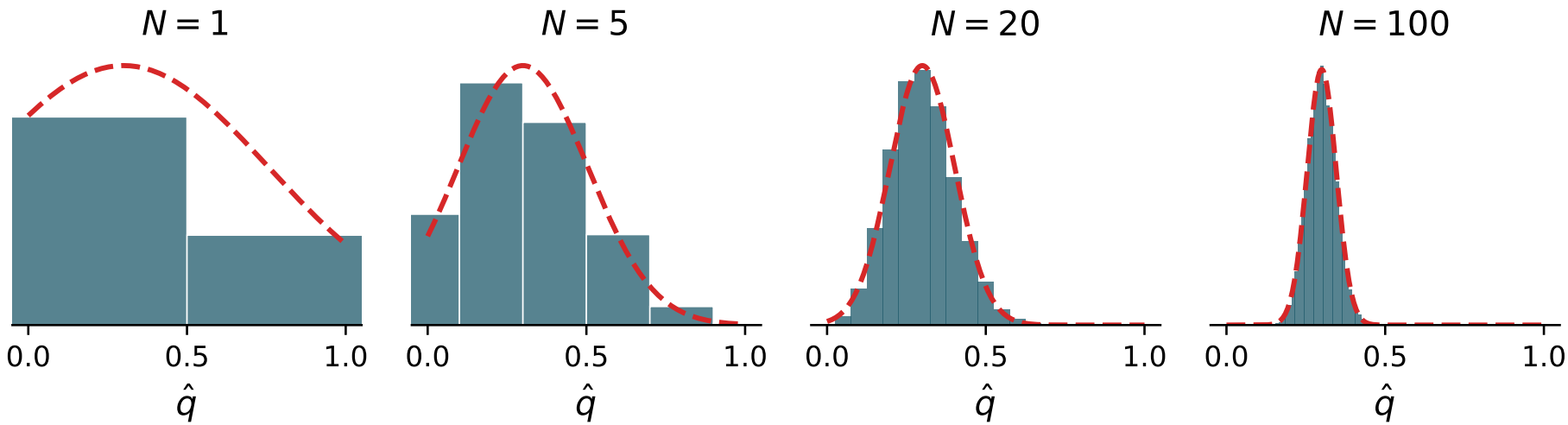
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Estimating q

Example

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$\hat{q}$ is not Normal, but by the CLT the Normal result carries over for large N, with $\sigma^2 = q(1-q)$:

$$\hat{q} \approx \text{Normal}\left(q,\ \frac{q(1-q)}{N}\right), \qquad \text{se}(\hat{q}) = \sqrt{\frac{q(1-q)}{N}}.$$